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Palaeomagnetism and Age of Bhander and Rewa Sandstones from India

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Summary

Palaeomagnetic measurements have been carried out on 38 oriented samples of sandstones belonging to the Bhander and Rewa series of Upper Vindhyan system, exposed near Saugar (23° 51′ N, 78° 45′ E), India. The mean remanent directions obtained after stability tests of alternating field demagnetization and thermal cleaning give Palaeomagnetic pole positions at 35° N, 138° W (Rewa sandstone) and 31·5° N, 161° W (Bhander sandstone). These pole positions fall very close to that for the Purple Sandstone of Lower Cambrian age from the Salt-Range, West Pakistan. This agreement helps in establishing the age of the Rewa and Bhander series which are devoid of any material suitable for radioactive dating.

Introduction

The Palaeomagnetic data obtained so far on Indian formations has been classified (Athavale *et al.* 1970) into the following three groups.

1. Late Mesozoic–Early Tertiary igneous activity represented by the Deccan, Rajmahal, Rajmahendri and Sylhet traps and their associated intrusives.

2. Sediments belonging to the Gondwana system and covering a span from Upper Carboniferous to Cretaceous.

3. Precambrian igneous and sedimentary formations.

Palaeomagnetic data on formations of lower Palaeozoic age could not be obtained so far, mainly because the peninsular shield area of India is devoid of any formations belonging to this age group. Palaeozoic formations are very well developed in the Himalayas, where they have been highly disturbed due to folding, faulting and overthrusts. The '*in situ*' position of the samples from this relatively unexplored region cannot be guaranteed, unless Palaeomagnetic sampling is preceded by detailed field geological studies. Another reason why Himalayan rocks are less than satisfactory for obtaining Palaeomagnetic data useful in reconstruction of Gondwanaland is that at least some portions of the Himalayan mountain system are made up of sediments deposited in the 'Tethys' sea and are believed to have been uplifted as a consequence of northward drift of the Indian shield, after its breaking away from the supercontinent of Gondwanaland.

The absence of suitable Palaeozoic formations has left a vital gap in the Palaeomagnetic data from India, from the point of view of obtaining a complete picture of the drift history of this subcontinent.

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The Vindhyan System

Amongst the Precambrian formations of India, the Vindhyan System is the youngest and most extensive both laterally as well as vertically. The sediments cover an area of approximately 180 000 sq. km and are represented by a maximum thickness of 4200 m of sandstones, shales and limestones. The system is mainly developed in Central India and the Jodhpur Sandstones occuring in the Rajasthan State of Western India are considered as probable equivalents of Upper Vindhyan formations (Pascoe 1959). The subdivisions of the main Vindhyan system (Misra 1969) are given in Table 1.

Crawford & Compston (1970) have carried out some work on the age of the Vindhyan system. They have obtained an age of 745 ± 10 My for Malani Rhyolites of Rajasthan, which are unconformably overlain by the Jodhpur Sandstone. Crawford & Compston (1970) have also obtained an age of 1140 ± 12 My for the kimberlitic pipe intrusive into Kaimur series, by using the Rb–Sr method. However, Tugarinov *et al.* (1965) have dated Glauconite specimens from the upper Kaimur series by the K–Ar method and have obtained an age of 910 My. The age value given by Crawford & Compston (1970) for the intrusive into Kaimur series therefore appears to be on the higher side by about 100 My, even after the 910 My value is corrected by about 15 per cent to 1045 My, as suggested by Hurely (1966), to account

Table 1

Stratigraphic succession of Vindhyan System

Upper Vindhyans		
Bhander Series 500 m	Upper Bhander Shales	
	Upper Bhander Limestones	
	Upper Bhander Sandstones	
	Sirbu Shales	
	Lower Bhander Sandstones	
	Lower Bhander Limestones	
	Ganurgarh Shales	
Rewa Series 150–300 m	Upper Rewa Sandstones	
	Jhiri Shales	
	Lower Rewa Sandstones	
	Panna Shales	
Kaimur Series 150–300 m	Upper Kaimur Sandstones (Dhandhraul Quartzites)	
	Upper Kaimur Conglomerates (Scarp Sandstones)	
	Bijaigarh Shales	
	Upper Quartzite	
	Silicified Shales	
	Lower Quartzites	
Lower Vindhyans		
Semri Series 300–1000 m	Rohtas Stage	
	Kheinjua Stage	Fawn Limestone
		Olive Shales
	Porcellanite Stage	
	Basal Stage	
		Kajrahat Limestone
		Conglomerates
	Bijawar Metamorphics	

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for the systematic error involved in K–Ar dating of Glauconite samples coming from Proterozoic rocks.

The Rewa and Bhander series occupy the upper part of the Vindhyan system and represent nearly half of its total thickness. They do not contain any material suitable for radioactive dating. On the basis of the prevalent geological opinion that these formations are coeval with the Jodhpur Sandstone which overlie the Malani Rhyolites (745 ± 10 My), Crawford & Compston (1970), suggest that these uppermost Vindhyans could as well be of lower Palaeozoic age, in view of their enormous thickness.

We report in this paper, the Palaeomagnetic results on sandstones from the upper part of the Bhander series, which is the youngest member of the Vindhyan system and on a site in sandstones from the upper part of Rewa series. They were chosen with the hope that they might yield some information regarding the position occupied by India during the Lower Palaeozoic.

Sampling details and palaeomagnetic results

Twenty-five oriented block samples of upper Bhander sandstones were collected from four sites located between Nohata (23° 41′ N, 79° 34′ E) and Katangi (23° 27′ N, 79° 48′ E) on the road from Saugar (23° 51′ N, 78° 45′ E) to Jabalpur in Madhya Pradesh. In addition 13 samples were collected from two quarries in the Vindhyan Sandstones exposed near Saugar and considered as belonging to the upper Rewa stage of the Vindhyan system (private communication from Professor W. D. West, Saugar University). All the beds sampled by us were almost horizontal, having very low dips.

Cores having diameter of 2·5 cm were drilled out from each oriented sample block and these oriented cores were sliced to obtain specimens having a height of approximately 2·5 cm. On an average, about four specimens were obtained from each block. The Natural Remanent Magnetization (NRM) of each specimen was measured on an Astatic Magnetometer having a sensitivity of 10^{-7} Oe cm^{-1} deflection. The direction and intensity of magnetization (Jo) were calculated in each case. The magnetic susceptibilities of the specimens were measured in an alternating field of 0·5 Oe, on an apparatus constructed after the design by Likhite & Radhakrishnamurty (1965). Table 2 gives the mean values and ranges of susceptibility and NRM intensity for specimens from each site. As can be seen from this table,

Table 2

Formation	Site No.	No. of samples	No. of specimens	NRM Intensity in cgs Units $\times 10^{-6}$ Mean value	Range	Susceptibility in cgs Units $\times 10^{-6}$ Mean value	Range
Upper Rewa Sandstone	1	6	30	94·2	4·9–200·0	2·8	2·2–3·6
Upper Rewa Sandstone	2	7	27	171·2	14·4–1005·5	2·7	0·7–3·8
Upper Bhander Sandstone	3	5	13	6·0	2·9–11·1	5·8	3·0–8·6
Upper Bhander Sandstone	4	7	22	3·1	1·2–6·5	9·4	7·9–15·8
Upper Bhander Sandstone	5	6	21	13·1	9·2–23·1	9·3	8·0–10·4
Upper Bhander Sandstone	6	7	25	5·8	3·0–12·0	10·1	8·0–14·1

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the samples are in general weakly magnetised and the NRM intensities of samples from one of the sites (No. 4) in the Bhander Sandstone are lower than those for the other sites by at least a factor of two.

In order to examine the stability of the NRM, one pilot specimen from each sample from a site was subjected to Alternating Field (A.F.) demagnetization in successively increasing peak field-strengths of 50, 100, 200, 300, 400, 600 and 800 Oe. The direction and intensity (J) of remanent magnetization was measured after each stage of demagnetization. The optimum alternating field value for batch cleaning of all the specimens from a site was decided on studying the pattern of vector migration and intensity decay, shown by the results obtained on the pilot specimens from each site.

This procedure is illustrated through Figs 1–4 in the case of specimens from site No. 2. A comparison of Fig. 4 with Fig. 1 shows, that the scatter in the directions of specimens from site No. 2 has been considerably reduced and the effect of secondary components has been removed as a consequence of A.F. demagnetization in peak-field of 600 Oe. Figs 5, 6 and 7 show the stereographic projections of stable remanent directions and the mean direction obtained by using the same procedure in the case of site Nos 3, 5 and 6 respectively.

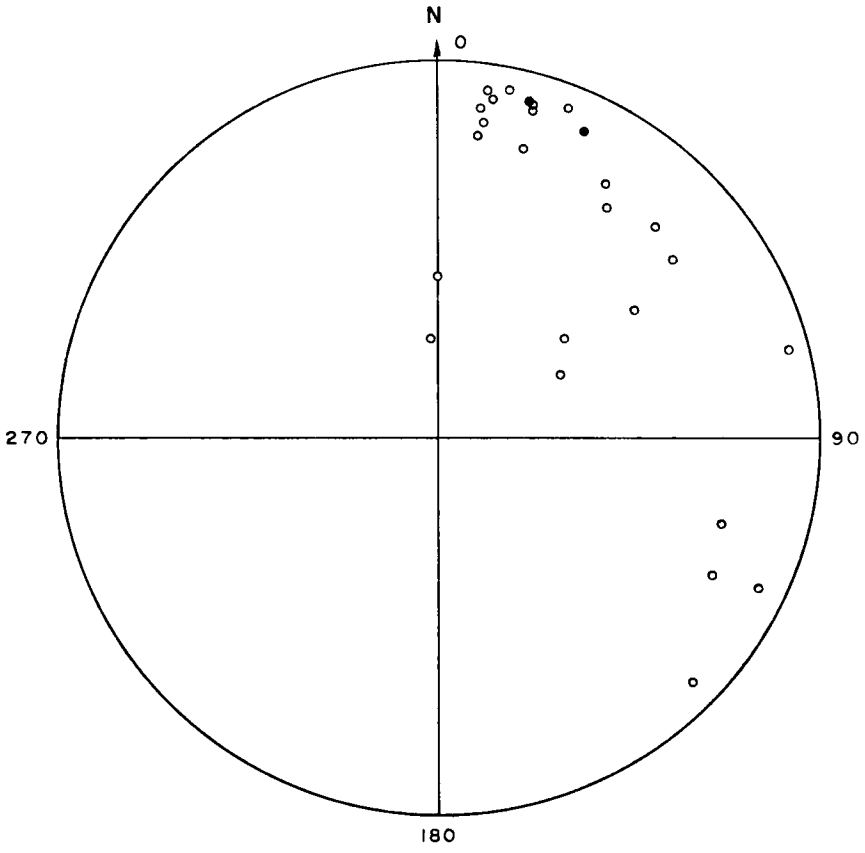


FIG. 1. NRM directions for Site No. 2 plotted on a stereographic projection; solid circles—lower hemisphere, open circles—upper hemisphere.

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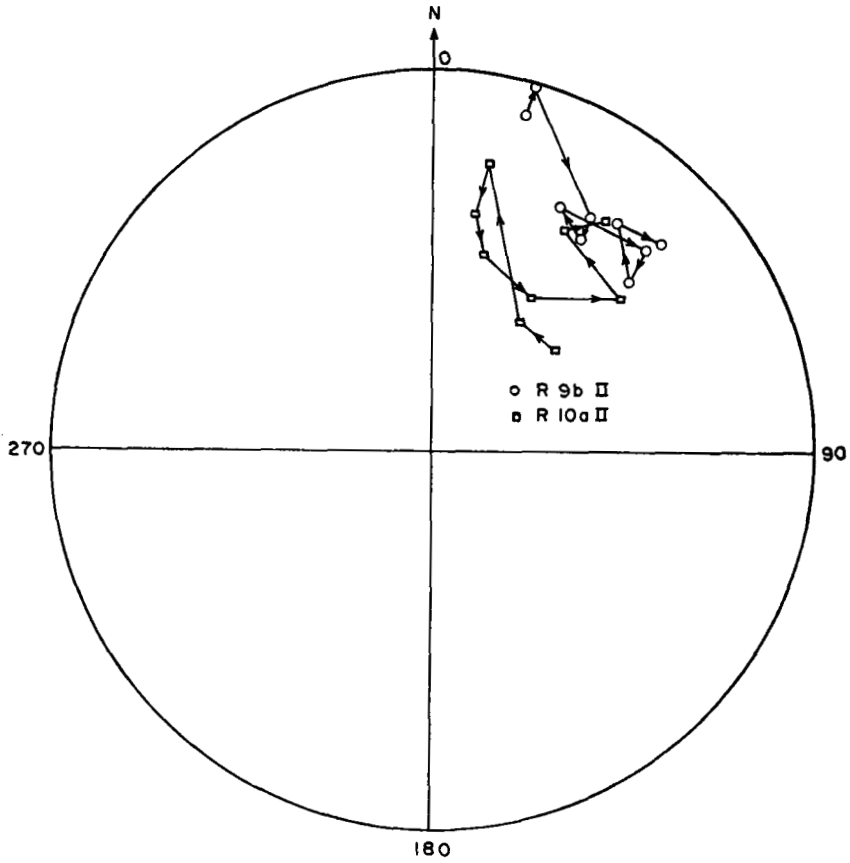


FIG. 2. Change in the direction of magnetization of two specimens (R9 bII and R10 aII) from Site No. 2 on A.F. demagnetization in successive steps of 50, 100, 200, 300, 400, 500, 600 and 800 Oe peak-field stereographic projection, upper hemisphere.

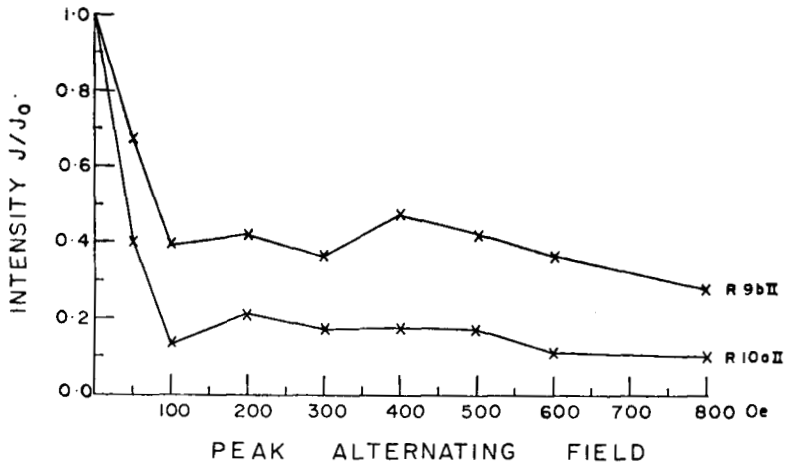


FIG. 3. Variation of normalized magnetic intensity (J/J_0) with peak alternating field-strength for two specimens (R9 bII and R10 aII) from Site No. 2.

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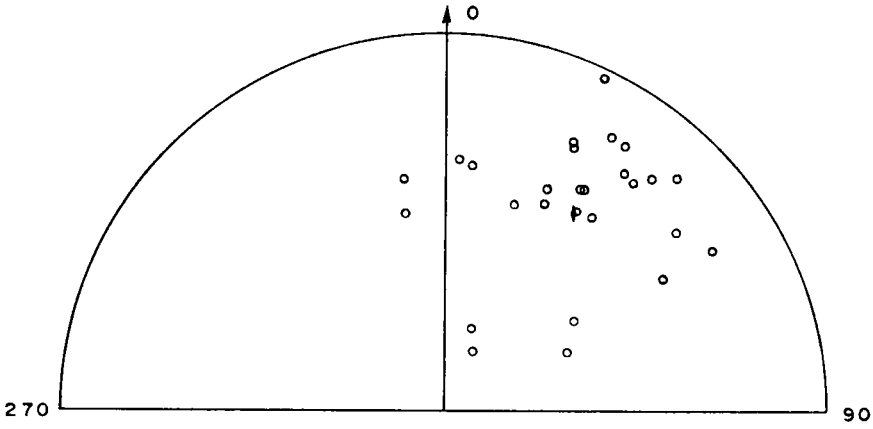
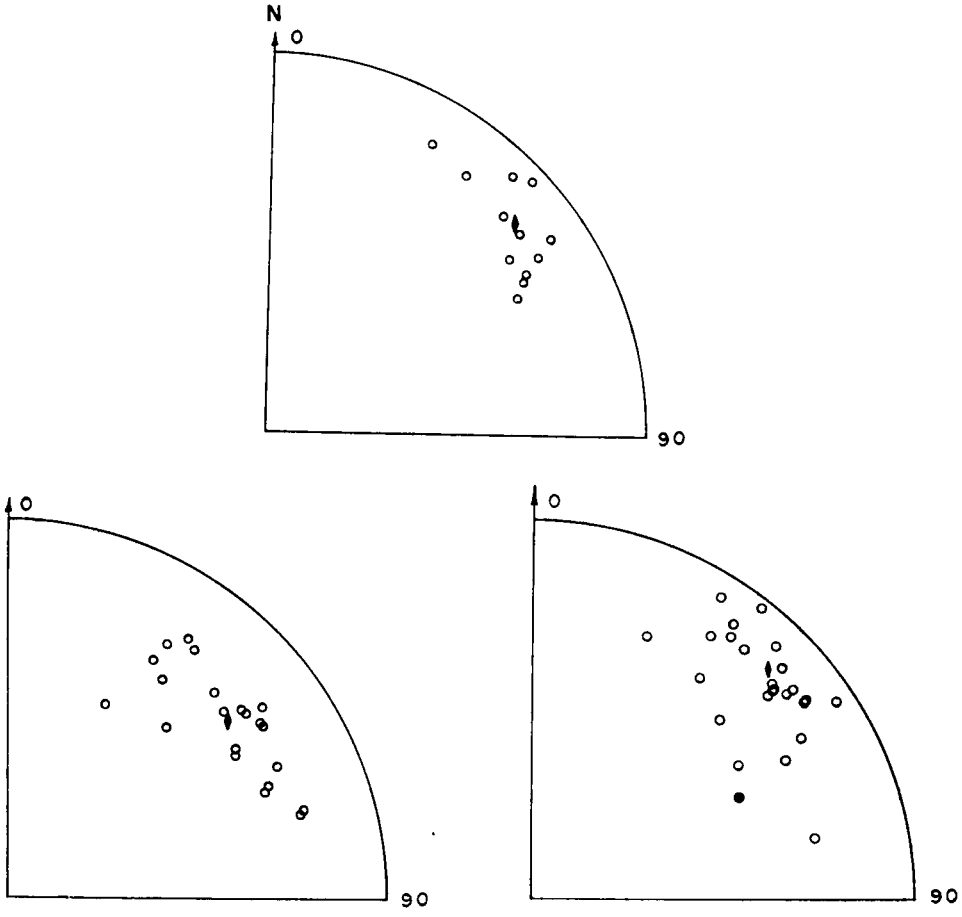


FIG. 4. The RM directions of specimens from Site No. 2 on demagnetization in 600 Oe, projection and symbols as in Fig. 1◆—mean direction.



FIGS 5, 6 and 7. The RM directions of specimens from Site Nos. 3, 5 and 6 respectively; projection and symbols as in Fig. 4.

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The intensity of remanent magnetization of specimens from one site in the Bhander Sandstone (No. 4) dropped below the limit of measurement sensitivity (1×10^{-7} Oe cm^{-1} deflection) of our magnetometer, on demagnetization in 200 Oe. The specimens from this site were not processed any further.

Although considerable reduction in the scatter of the directions was observed in the case of three Bhander sites (Nos 3, 5 and 6) and one of the Rewa sites (No. 2) on A.F. cleaning, this procedure was found to be ineffective in reducing the scatter in the directions of specimens from the other Rewa site (No. 1) as can be seen from Figs 8 and 9. Specimens from this site were therefore subjected to thermal cleaning in an apparatus specially designed for thermal demagnetization of sedimentary rocks. This apparatus consists of six pairs of Helmholtz coils, which compensate the Earth's magnetic field up to 0·1 per cent of its value at the centre of the system, over a spherical volume of 10-cm diameter. The specimens are heated in a non-magnetic furnace and the apparatus gives satisfactory results on thermal demagnetization of sandstones up to 600 °C (Athavale 1969). The thermal cleaning treatment did not help in reducing the scatter in directions measured on heating up to 350 °C and cooling in zero field (Fig. 10). The remanent intensity values however dropped from 1 to 5 per cent of the NRM value as a consequence of heating up to this temperature. Thermal cleaning at higher temperatures was therefore considered as unwarranted. The results from this site were omitted from further analysis.

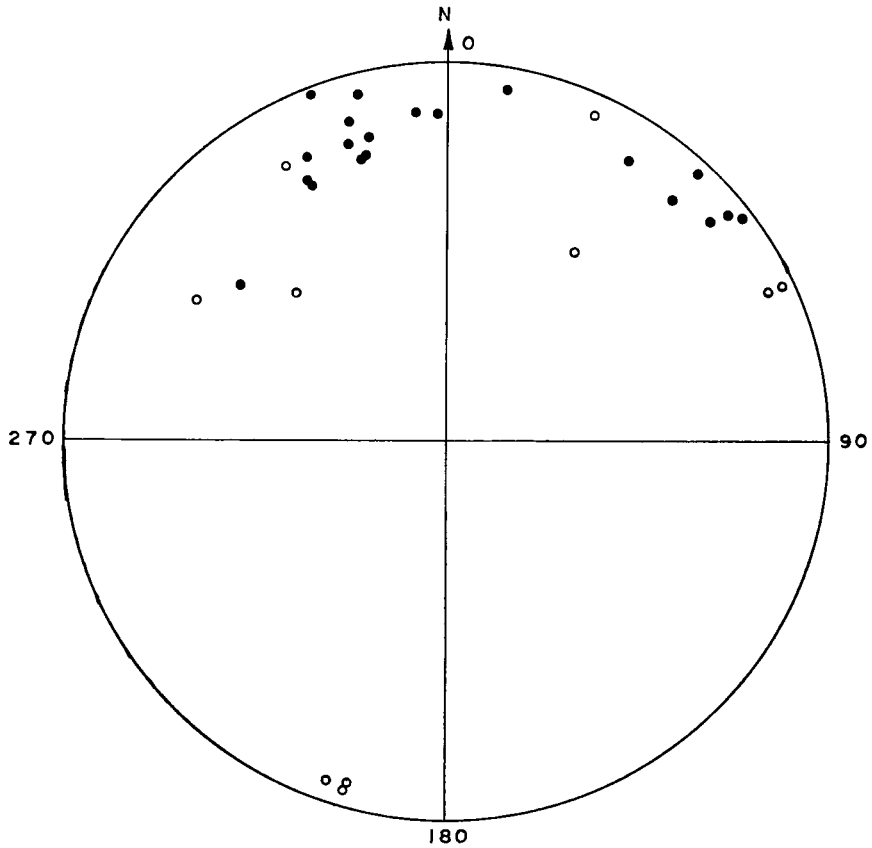


FIG. 8. The NRM directions for Site No. 1. Projection and symbols as in Fig. 1.

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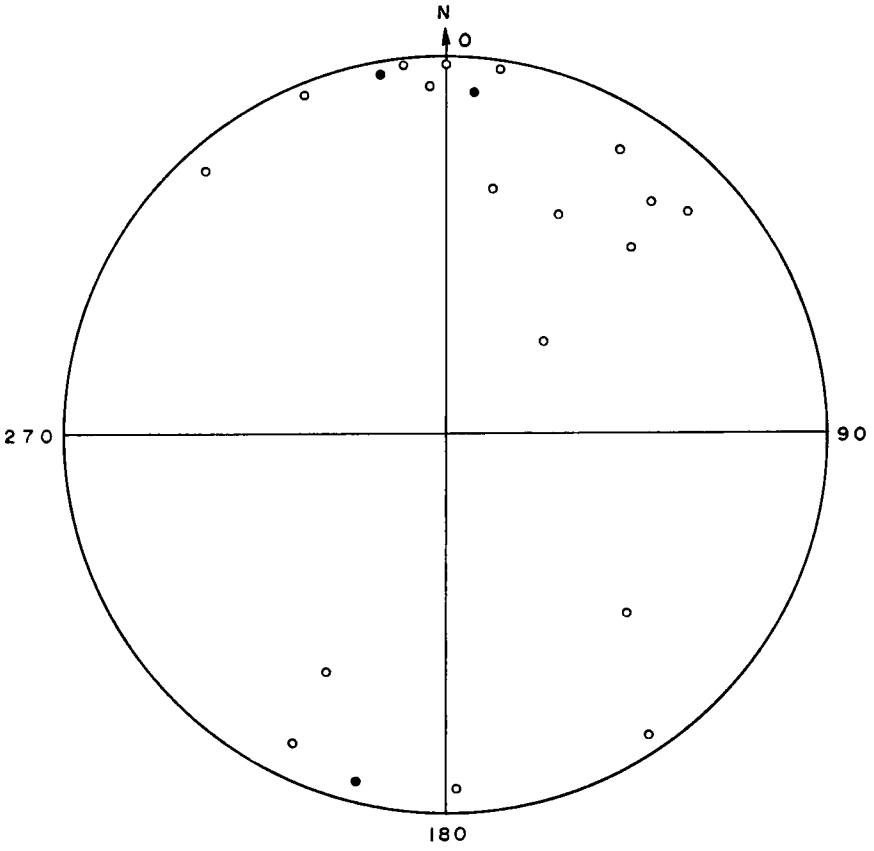


FIG. 9. The scatter in RM vectors of specimens from Site No. 1 after demagnetization in a peak field of 600 Oe. Stereographic projection.

Analysis of data

The stable remanent directions for specimens from each site were given unit weight for obtaining mean directions for samples (Fisher 1953). The samples were similarly given unit weight for obtaining the mean stable remanent direction for the site. In Table 3 are presented the mean stable remanent magnetization directions for one upper Rewa and three upper Bhander sites. Also given in the table are the radius of circle of confidence and the precision parameter calculated in each case, by using Fisher's (1953) method. The Palaeolatitudes of a central Indian town-Nagpur (21° N, 79° E)-calculated from these directions, are also given in the Table.

In Table 4 are given the pole positions obtained for the single site in upper Rewa Sandstone and the mean pole position for three sites in Upper Bhander Sandstone. Also given in the Table are the Palaeomagnetic pole positions for Malani Rhyolite (Athavale, Radhakrishnamurty & Sahasrabudhe 1963), Kaimur quartzites (Sahasrabudhe & Misra 1966) and Purple Sandstone from Salt-Range (McElhinny 1970).

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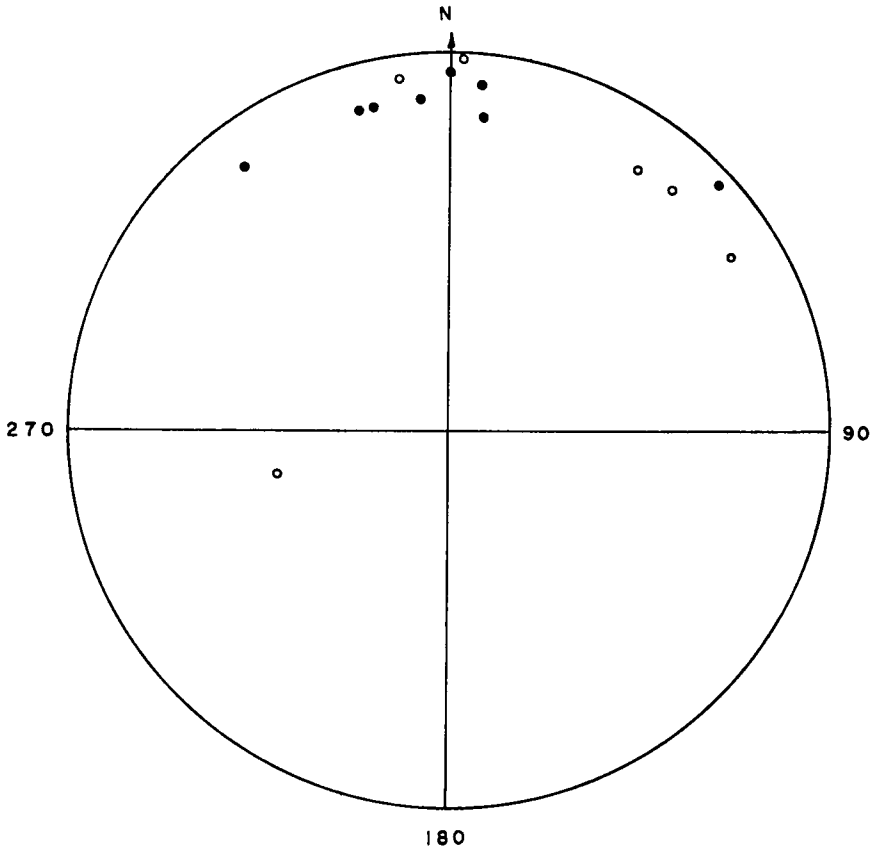


FIG. 10. The scatter in RM directions of specimens from Site No. 1 after thermal demagnetization up to 350°C in zero-field.

Table 3

Formation	Site No.	Location	No. of samples	No. of specimens	Mean direction D	Mean direction I	α95	K	Palaeolatitudes of Nagpur
Rewa Sandstone	2	23·8° N 78·9° E	7	27	32°	−37°	13·7°	15	24·0° S
Bhander Sandstone	3	23·6° N 79·7° E	5	12	49°	−16°	8·3°	57	10·5° S
Bhander Sandstone	5	23·7° N 79·6° E	6	21	51°	−26°	8·5°	45	15·8° S
Bhander Sandstone	6	23·7° N 79·6° E	7	25	46°	−15°	6·1°	75	9·8° S
Mean Bhander Sandstone			18	58	49°	−19°	5·7°	200	12·0° S

Table 4

Formation	N. pole position	δm	δP	Reference
Rewa Sandstone	35·0° N 138·0° W	15·9°	9·6°	This paper
Upper Bhander Sandstone (mean)	31·5° N 161·0° W	5·9°	3·1°	This paper
Purple Sandstone Salt-Range	28·0° N 148·0° W	13·9°	7·5°	McElhinny 1970
Kaimur Quartzites	82·0° N 103·0° W	6·7°	3·7°	Sahasrabudhe & Misra 1966
Malani Rhyolite	78·0° N 45·0° E	14·4°	10·3°	Athavale *et al.* 1963

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Discussion

Fig. 11 is a stereographic projection of Palaeomagnetic pole positions for the formations listed in Table 4. The pole position for the Cambrian Purple Sandstone from the Salt-Range lies between those for Upper Rewa Sandstone and Upper Bhander Sandstone and the three are closely grouped. Palaeomagnetic data thus supports the observation made by Fox (1929) that the prevalence of red colour and occurrence of slat-pseudomorph beds in the Upper Vindhyan Sandstones as well as in the sandstones from the Salt-Range suggest that they were formed under similar climatic conditions and are of the same age. It is also possible to deduce from Fig. 11 that the upper Bhander Sandstones are slightly younger and the upper Rewa Sandstone underlying them are slightly older than the Purple Sandstone.

According to McElhinny (1970), on the basis of fossil records, the age of the Purple Sandstone can be reliably fixed as Early Cambrian. Assuming that the Palaeomagnetic Pole position given by McElhinny (1970) for this formation from West Pakistan, represents the Lower Cambrian Palaeomagnetic Pole for the entire Indian subcontinent and that there has not been any relative movement between the Salt-Range region and the main Indian shield during geological times, it can be

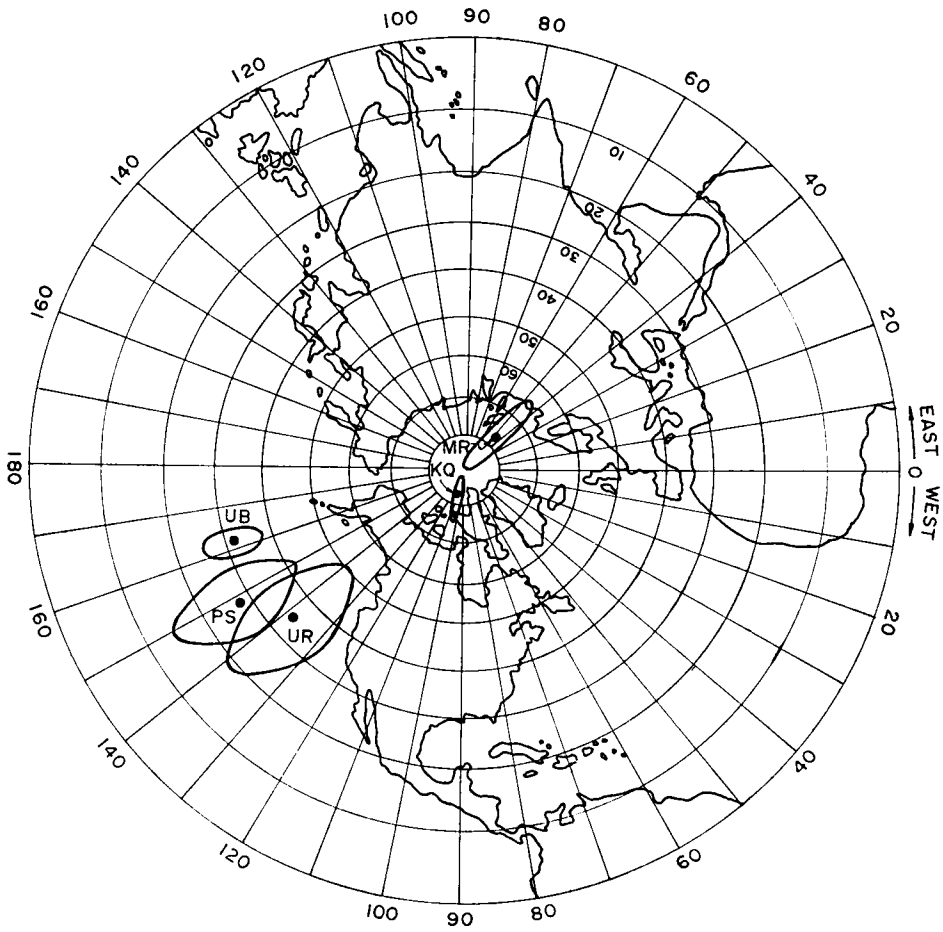


FIG. 11. Palaeomagnetic pole positions; UR = Upper Rewa Sandstone, UB = Upper Bhander Sandstone, PS = Purple Sandstone, KQ = Kaimur Quartzites and MR = Malani Rhyolites. The oval of confidence is shown in each case

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suggested that the upper Rewa series is of Late Precambrian or earliest Cambrian age and that the upper Bhander series belongs to the Middle Cambrian. On this basis, the upper limit of Vindhyan sedimentation can be fixed at around 550 My on the geological time scale (Van Eysinga 1971).

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